

Even π -atom count

Structural precondition:
Kekulé matching requires
even-vertex graph

necessary

$$|\text{Aut}(\text{G})| \leq 2$$

Symmetry gate:
low symmetry permits
bond inequivalence

necessary

$$K/|\text{Aut}(\text{G})| \geq 5.0$$

Localization index:
matching multiplicity
survives symmetry discount



required

Bay/Fjord region

Geometric gate:
diol-epoxide pathway
is structurally accessible

→ Not sufficient
alone

→ **CARCINOGENIC**

Low symmetry ($|\text{Aut}| \leq 2$) is necessary but not sufficient. $K/|\text{Aut}(\text{G})| \geq 5.0$ adds graded localization potential. Bay/fjord geometry provides the mechanistic exit to the diol-epoxide pathway.